

CE3420

Assignment

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The Python Standard Library

```
In [2]: """ Question
Find the coordinates of 2 cities near you and calculate the distance between them """

import math

def haversine_distance(origin, destination):
    lat1, lon1 = origin
    lat2, lon2 = destination
    radius = 6371000
    dlat = math.radians(lat2-lat1)
    dlon = math.radians(lon2-lon1)
    a = math.sin(dlat/2) * math.sin(dlat/2) + math.cos(math.radians(lat1))
        * math.cos(math.radians(lat2)) * math.sin(dlon/2) * math.sin(dlon/2)
    c = 2 * math.atan2(math.sqrt(a), math.sqrt(1-a))
    distance = radius * c
    return distance

hyderabad = (17.4065, 78.4772)
bengaluru = (12.9629, 77.5775)

distance = haversine_distance(hyderabad, bengaluru)
print(f"Distance: {distance/1000} 'km'")
```

Distance: 503.44469133342545 'km'

Third-party Modules

```
In [ ]: """ Question
Repeat the distance calculation exercise from the previous module but per-
form it using the geopy module """

%pip install geopy
from geopy import distance

hyderabad = (17.4065, 78.4772)
bengaluru = (12.9629, 77.5775)

straight_line_distance = distance.great_circle(hyderabad, bengaluru)
ellipsoid_distance = distance.geodesic(hyderabad, bengaluru, ellipsoid='WGS84')

print(f"straight_line_distance: {straight_line_distance}")
print(f"ellipsoid_distance: {ellipsoid_distance}")
```

Defaulting to user installation because normal site-packages is not writeable

Collecting geopy

Downloading geopy-2.4.1-py3-none-any.whl (125 kB)

|██| 125 kB 3.0 MB/s eta 0:00:01

Collecting geographiclib<3,>=1.52

Downloading geographiclib-2.1-py3-none-any.whl (40 kB)

|██| 40 kB 10.7 MB/s eta 0:00:01

Installing collected packages: geographiclib, geopy

Successfully installed geographiclib-2.1 geopy-2.4.1

WARNING: You are using pip version 21.2.4; however, version 26.0 is available.

You should consider upgrading via the '/Library/Developer/CommandLineTools/usr/bin/python3 -m pip install --upgrade pip' command.

Note: you may need to restart the kernel to use updated packages.

straight_line_distance: 503.4454025251104 km

ellipsoid_distance: 501.09780212195164 km

Using Web APIs

```
In [6]: """ Question 1
Replace the ORS_API_KEY with your own key in the code below. Change the c
"""

%pip install requests
import requests

ORS_API_KEY = 'eyJvcmdiOiI1YjNjZTM1OTc4NTExMTAwMDFjZjYyNDgiLCJpZCI6IjdiND

hyderabad = (17.4065, 78.4772)
bengaluru = (12.9629, 77.5775)

parameters = {
    'api_key': ORS_API_KEY,
    'start' : '{}{}'.format(hyderabad[1], hyderabad[0]),
    'end' : '{}{}'.format(bengaluru[1], bengaluru[0])
}

response = requests.get(
    'https://api.openrouteservice.org/v2/directions/driving-car', params=

if response.status_code == 200:
    print('Request successful.')
    data = response.json()
else:
    print('Request failed.')
    print('Reason', response.text)

data = response.json()

summary = data['features'][0]['properties']['summary']
print(summary)
```

```

Defaulting to user installation because normal site-packages is not writeable
Collecting requests
  Downloading requests-2.32.5-py3-none-any.whl (64 kB)
    |██████████████████████████████████████| 64 kB 3.1 MB/s eta 0:00:01
Collecting idna<4,>=2.5
  Downloading idna-3.11-py3-none-any.whl (71 kB)
    |██████████████████████████████████████| 71 kB 4.7 MB/s eta 0:00:01
Collecting certifi>=2017.4.17
  Downloading certifi-2026.1.4-py3-none-any.whl (152 kB)
    |██████████████████████████████████████| 152 kB 10.7 MB/s eta 0:00:01
Collecting urllib3<3,>=1.21.1
  Downloading urllib3-2.6.3-py3-none-any.whl (131 kB)
    |██████████████████████████████████████| 131 kB 4.5 MB/s eta 0:00:01
Collecting charset_normalizer<4,>=2
  Downloading charset_normalizer-3.4.4-cp39-cp39-macosx_10_9_universal2.whl (209 kB)
    |██████████████████████████████████████| 209 kB 5.5 MB/s eta 0:00:01
Installing collected packages: urllib3, idna, charset-normalizer, certifi, requests
Successfully installed certifi-2026.1.4 charset-normalizer-3.4.4 idna-3.11 requests-2.32.5 urllib3-2.6.3
WARNING: You are using pip version 21.2.4; however, version 26.0 is available.
You should consider upgrading via the '/Library/Developer/CommandLineTools/usr/bin/python3 -m pip install --upgrade pip' command.
Note: you may need to restart the kernel to use updated packages.
/Users/17kaushalsingh/Library/Python/3.9/lib/python/site-packages/urllib3/__init__.py:35: NotOpenSSLWarning: urllib3 v2 only supports OpenSSL 1.1.1+, currently the 'ssl' module is compiled with 'LibreSSL 2.8.3'. See: https://github.com/urllib3/urllib3/issues/3020
  warnings.warn(
Request successful.
{'distance': 569048.9, 'duration': 27527.6}

```

```

In [7]: """ Question 2
Below cell contains a dictionary with 3 destination cities and their coordinates
"""

import requests
import time

ORS_API_KEY = 'eyJvcnVjcmciOiI1YjNjZTM1OTc4NTExMTAwMDFjZjZjYyNDgiLCJpZCI6IjdiND'

def get_driving_distance(source_coordinates, dest_coordinates):
    parameters = {
        'api_key': ORS_API_KEY,
        'start' : '{},{ {}'.format(source_coordinates[1], source_coordinates[0])
        'end' : '{},{ {}'.format(dest_coordinates[1], dest_coordinates[0])
    }

    response = requests.get(
        'https://api.openrouteservice.org/v2/directions/driving-car', par

    if response.status_code == 200:
        data = response.json()
        summary = data['features'][0]['properties']['summary']
        distance = summary['distance']
        return distance/1000

```

```

else:
    print('Request failed.')
    print('Reason', response.text)
    return -9999

san_francisco = (37.7749, -122.4194)

destination_cities = {
    'Los Angeles': (34.0522, -118.2437),
    'Boston': (42.3601, -71.0589),
    'Atlanta': (33.7490, -84.3880)
}

for city, coordinates in destination_cities.items():
    distance = get_driving_distance(san_francisco, coordinates)
    print(f"Driving distance from San Francisco to {city} is {distance} km")
    time.sleep(2)

```

Driving distance from San Francisco to Los Angeles is 615.5596999999999 km
 Driving distance from San Francisco to Boston is 4991.4701 km
 Driving distance from San Francisco to Atlanta is 3976.3255 km

Review of Common Python Errors, Reading Files

```

In [8]: """ Question
Count the number of lines in the file and print the total count.
You can adapt the for-loop above to count the total number lines in the f

# Add code to open the file and print the total number of lines
# Hint: You do not need to print the line, just increment the count
# and print the value of the variable once the loop is finished.

"""

import os
data_pkg_path = 'python_foundation/data'
filename = 'worldcities.csv'
path = os.path.join(data_pkg_path, filename)

line_count = 0
with open(path, 'r', encoding='utf-8') as file:
    for line in file:
        line_count += 1
print(f"Total number of lines in the file: {line_count}")

```

Total number of lines in the file: 15494

Reading CSV Files

```

In [17]: """ Question
Replace the home_city and home_country variables with your own home city
"""

import csv
import os
from geopy import distance

data_pkg_path = 'python_foundation/data'

```

```

filename = 'worldcities.csv'
path = os.path.join(data_pkg_path, filename)

output_dir = 'Assignment2'
output_filename = 'cities_distance.csv'
output_path = os.path.join(output_dir, output_filename)

if not os.path.exists(output_dir):
    os.mkdir(output_dir)

home_city = 'Hyderabad'
home_country = 'India'

home_city_coordinates = (17.4065, 78.4772)

with open(path, 'r', encoding='utf-8') as input_file:
    csv_reader = csv.DictReader(input_file)
    for row in csv_reader:
        if row['city_ascii'] == home_city and row['country'] == home_coun
            home_city_coordinates = (row['lat'], row['lng'])
            break

# Write distances to CSV
with open(output_path, mode='w', newline='', encoding='utf-8') as output_
    fieldnames = ['city', 'distance_from_home']
    csv_writer = csv.DictWriter(output_file, fieldnames=fieldnames)
    csv_writer.writeheader()

    with open(path, 'r', encoding='utf-8') as input_file:
        csv_reader = csv.DictReader(input_file)
        for row in csv_reader:
            if row['country'] == home_country and row['city_ascii'] != ho
                city_coordinates = (row['lat'], row['lng'])
                city_distance = distance.geodesic(
                    city_coordinates,
                    home_city_coordinates
                ).km

                csv_writer.writerow({
                    'city': row['city_ascii'],
                    'distance_from_home': city_distance
                })

print(f"Successfully written output file at {output_path}")

```

Successfully written output file at Assignment2/cities_distance.csv

Working with Pandas

```

In [22]: """ Question
You will notice that the output file contains a row with the home_city as
Modify the filtered dataframe to remove this row and write the results to

Hint: Use the Boolean filtering method we learnt earlier to select rows t
"""

import os
from geopy import distance

```

```

import pandas as pd

def calculate_distance(row):
    city_coordinates = (row['lat'], row['lng'])
    return distance.geodesic(city_coordinates, home_city_coordinates).km

data_pkg_path = 'python_foundation/data'
filename = 'worldcities.csv'
path = os.path.join(data_pkg_path, filename)
df = pd.read_csv(path)

home_country = 'India'
home_city = 'Bengaluru'

country_df = df[df['country'] == home_country].copy()
filtered = country_df[country_df['city_ascii'] == home_city]
home_city_coordinates = (filtered.iloc[0]['lat'], filtered.iloc[0]['lng'])

country_df['distance'] = country_df.apply(calculate_distance, axis=1)

filtered = country_df[['city_ascii', 'distance']]
filtered = filtered.rename(columns={'city_ascii': 'city'})

filtered = filtered[filtered['city'] != home_city]

filtered

```

Out[22]:

	city	distance
3	Mumbai	837.185709
5	Delhi	1738.638856
7	Kolkata	1552.637823
34	Chennai	295.340107
41	Hyderabad	500.047729
...	...	...
7305	Karur	230.567496
7441	Jorhat	2312.574457
7583	Sopur	2383.154991
7681	Tezpur	2195.314732
9384	Diu	1106.693243

211 rows × 2 columns